Personal Book Universe Tracker

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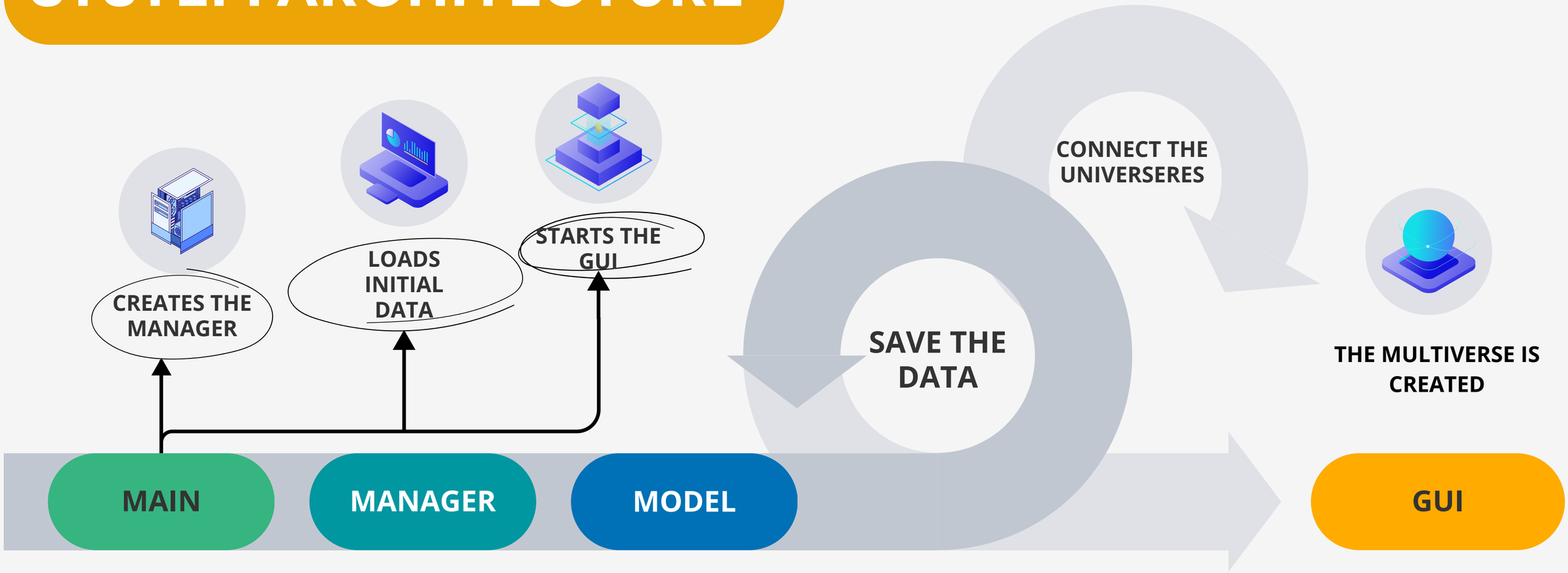
INTRODUCTION

- Java-based application
- Manages fictional universes, series, books, characters
- Helps organize and explore relationships

OBJECTIVES

- Implement OOP concepts
- Design hierarchical system
- Build GUI using Swing
- Manage book & character data

SYSTEM ARCHITECTURE



ENTRY POINT

RUNS EVERYTHING IN ORDER

LIBRARY MANAGER

Handles logic and Loads data

UNIVERSES
SERIES
BOOKS
CHARACTERS

USER INTERACTION AND VIEW

OOP Concepts Used

Encapsulation:
Private data + getters/setters

Polymorphism:
Book reference for Novel

Abstraction:
Manager & GUI hide logic

Inheritance:
Novel extends Book

FEATURES

- Organized structure
- GUI interaction
- Preloaded data
- Modular design

ERRORS OCCURED

- GUI layout issues
- Input handling
- Debugging errors
- Designing relationships

EXISTING LIMITATIONS

- No database
- Basic UI
- Limited search
- No data persistence
- No access control

Future Roadmap

ADD DATABASE

FILE STORAGE

ADVANCED SEARCH

SCALE & MANUFACTURING OPTIMIZATION

LEARNING OUTCOMES

- OOP concepts
- Swing GUI
- Debugging
- System design

CONCLUSION

Successfully implemented OOP + GUI to build a structured and scalable system. And now I have a good foundation for future upgrades.

Thank You

FOR YOUR TIME